



## Karachi AQI Hub

AI-Powered Air Quality Forecasting & Health Intelligence

### System Status

Champion Model: Gradient Boosting

Test R<sup>2</sup> Score: 0.9113

Test RMSE: 2.39 µg/m<sup>3</sup>

### Hopworks Feature Store

#### Feature Group:

karachi\_aqi\_features (v1)

#### Selected Features: 18 Predictors

Sync Mode: Hourly GitHub Actions Cron

Refresh Live Data

Built for 10 Pearls Data Science Internship • Karachi, Pakistan

## Karachi Air Quality Index & 72-Hour Forecaster

Karachi, Pakistan • Real-Time Satellite & Atmospheric Telemetry • Updated: Thursday, August 27, 2026 • 09:00 AM UTC

LIVE AQI STATUS: MODERATE



PM10 Coarse Dust

62.4 µg/m³

Temperature

29.1 °C

Relative Humidity

70.0 %

Wind Speed

19.4 km/h

Carbon Monoxide (CO)

193 µg/m³

Surface Pressure

1002 hPa

**Health Advisory & Action Plan:** Air quality is acceptable; moderate health concern for sensitive individuals.

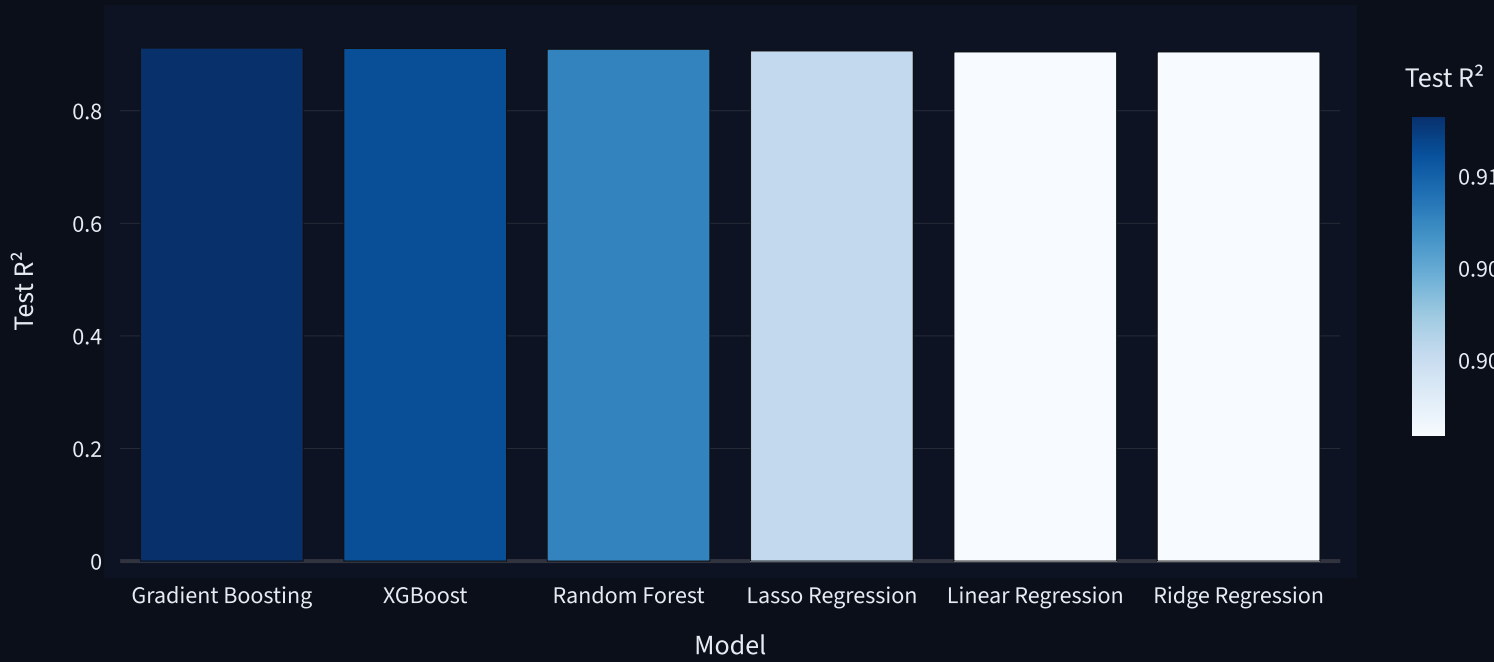
72-Hour (3-Day) AQI Forecast | Exploratory Data Analysis & Trends | SHAP Explainability & Factor Insights | Scenario Simulator Sandbox | Model Leaderboard & MLOps Registry

### Model Evaluation Benchmark & MLOps Registry

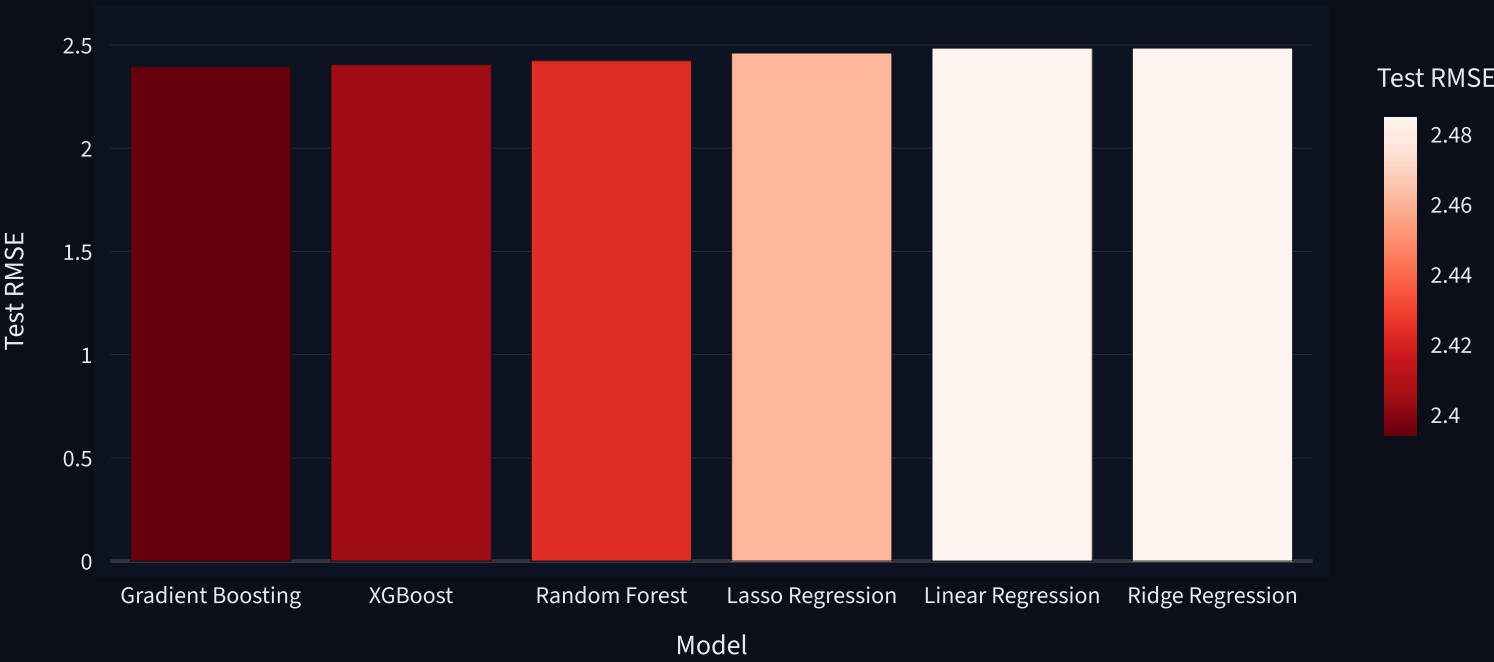
Rigorous multi-model performance comparison on chronological hold-out test set (3,427 hours).

|   | Model             | Test RMSE | Test MAE | Test R <sup>2</sup> | Train R <sup>2</sup> | Overfit Gap | Status     |
|---|-------------------|-----------|----------|---------------------|----------------------|-------------|------------|
| 0 | Gradient Boosting | 2.3945    | 1.4989   | 0.9113              | 0.9552               | 0.0439      | 🏆 Champion |
| 1 | XGBoost           | 2.4047    | 1.4891   | 0.9105              | 0.9711               | 0.0606      | Candidate  |
| 2 | Random Forest     | 2.4235    | 1.5248   | 0.9091              | 0.9789               | 0.0698      | Candidate  |
| 3 | Lasso Regression  | 2.4613    | 1.5567   | 0.9062              | 0.9271               | 0.0209      | Baseline   |
| 4 | Linear Regression | 2.4848    | 1.5849   | 0.9044              | 0.9271               | 0.0227      | Baseline   |
| 5 | Ridge Regression  | 2.4851    | 1.5851   | 0.9044              | 0.9271               | 0.0227      | Baseline   |

#### Test R<sup>2</sup> Score Comparison



#### Test RMSE Error Comparison (Lower is Better)



### Hopworks Feature Store Architecture

- Feature Group: karachi\_aqi\_features (v1)
- Primary Key: ['datetime', 'city']
- Event Time: datetime
- Features (18 Selected):
  - Lagged Features: pm25\_lag\_24, pm25\_lag\_48, pm25\_lag\_72
  - Rolling Features: pm25\_rolld\_mean\_24, pm25\_rolld\_std\_24, pm25\_rolld\_mean\_72, pm25\_rolld\_std\_72
  - Meteorology & Interactions: temp\_humidity, wind\_pm25, month\_cos
  - Pollutants: PM2.5, PM10, CO, SO2, NO2

- Model Registry: Local Versioned + Hopworks
- Champion Artifact: models/best\_model.pkl
- Metadata: models/best\_model\_metadata.json
- Retraining Cadence: Daily via GitHub Actions / Airflow
- Inference Cadence: Hourly automated streaming